VitalRide

A wearable vest and helmet to help **delivery riders** prevent **heatstroke** while working in hot weather

Wearable Devices /
Physiological Data monitoring/

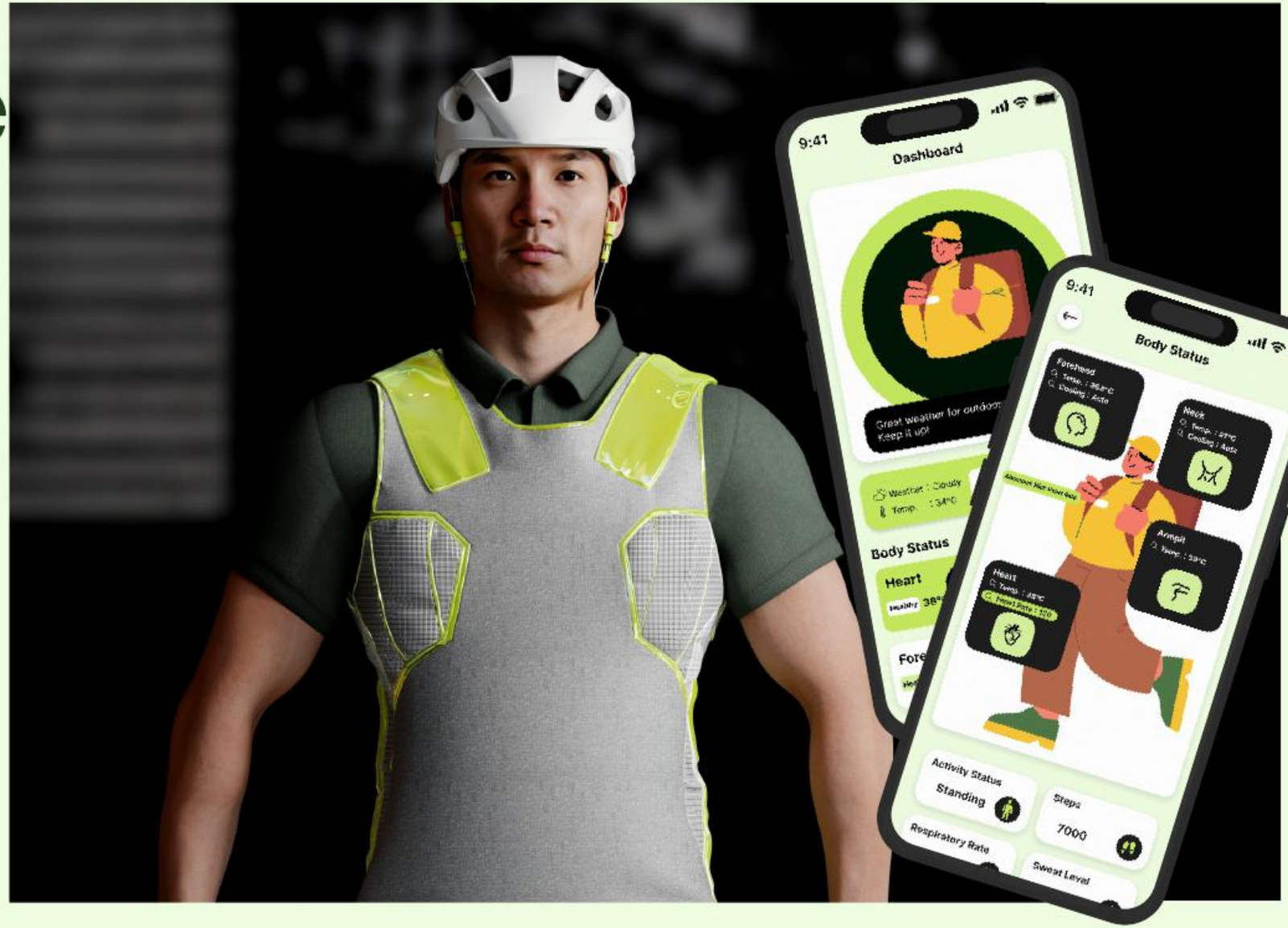
Service Design | UIUX Design | Individual Project

Project
01
02
03
04
05

Brief

The societal issue I identified is delivery riders' susceptibility to heatstroke during high temperatures, **a condition that can prove life-threatening in severe cases.** Through field research and desk studies, I pinpointed the core pain points: **riders lacked equipment for objectively monitoring physiological data** and **possessed insufficient awareness of heatstroke prevention.** My solution involved designing smart wearable devices to monitor vital signs, with an accompanying app delivering timely alerts, thereby **mitigating the threat of heatstroke for delivery personnel**.

Duration: 2025/06 - 2025/10



Insperation Collage

It was an early summer day when I first truly noticed delivery riders gathered in the cool refuge of a bubble tea shop. North China's summers are relentless—without such places, where could they escape the heat? That moment sparked my wish to design a product that could help riders monitor their physical condition with scientific accuracy. Instead of relying on sheer endurance until collapse, they could better safeguard their health and work more safely.



Background Research

What is Heatstroke?

A **life-threatening syndrome** characterized by a core body temperature exceeding 40°C and central nervous system abnormalities, accompanied by multi-organ damage.

Classic Heat Stroke



Patient population
Individuals with weakened constitutions

Causes
Prolonged exposure to high temperatures

Exertional Heat Stroke



Patient population
Individuals engaged in high-intensity labor under high temperatures

Causes
Excessive heat production coupled with **inadequate heat dissipation**

Exertional Heat Stroke

Case Study
Location: Hangzhou **Incident: Dizziness, Vomiting**

Aqiang, a delivery rider , said he had suffered heatstroke **4 times over the past 11 days.** His symptoms included **loss of consciousness, convulsions, confusion, vomiting, and high fever.**

Statement :

One midday (35°C) in early July, after delivering a meal, he felt **dizzy and lightheaded**, his limbs went limp and calves trembled violently, threatening to buckle him to his knees. Afterward, he **vomited** violently.



Hazards

- Accelerated **heart rate**
- Decreased blood pressure
- Abnormal liver function
- Skin is **hot to the touch**
- Coma
- Confusion

Causes



Exertional heat stroke happens when hard exercise in hot weather makes the body overheat.

Core Metrics



- **Core Body Temperature >40°C** is typically considered the threshold
- **Heart rate >100 bpm** is commonly observed
- Heatstroke patients exhibit **rapid breathing or even hyperventilation**

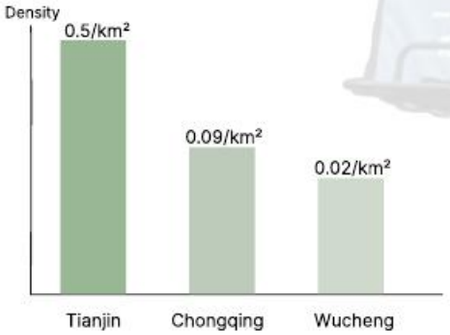
Critical Temperature of Organs

Degree of danger

Core Body	Above 38.5°C	Above 40°C	Above 42°C
Skin	Above 38°C	Above 41°C	Above 42°C
Heart	Above 39°C	Above 41°C	Above 42°C

Current Solutions

Diagram of Union Service Station Density in 3 Cities



Functions



Shortcomings

The density is too low for delivery personnel to receive timely support.

SURVEY RESEARCH

Interview Outline

What are your typical daily **working hours** for food delivery?
How do you cope with **extreme heat and intense sun exposure** during summer?
What moments during food delivery make it unbearable for you, or even put you **at risk of heatstroke**?

User Interview

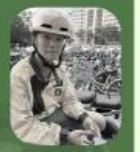
Heatstroke Prevention Or Making Money



Male, 38
11 hours

Closer to the merchant means faster pickup. **Handle the sun and the order's yours**; if not, someone else will take it.

Slow Speeds Intensify the Heat



Male, 19
10 hours

The hardest parts—walking in heat and climbing stairs—are **unavoidable** in every delivery.

Mental Stress Exacerbates Heatstroke



Male, 19
9 hours

Stuck in traffic, I feel anxious and **the heat worsens**. Rushing to meet delivery times under the blazing sun often **leaves me dizzy.**

Lack of Willingness to Prevent Heatstroke



Male, 35
13 hours

I have **no special way** to beat the heat—I just endure it. I'm not afraid of the sun since **I'm used to outdoor work.**

INSIGHTS AND INSPERATIONS

1. Low awareness of heatstroke and its dangers
2. Existing urban facilities are too sparse to provide timely support
3. Lack of real-time health data tracking

Field Research / Photos

01 Rest in the shade → 02 Delivery by e-bike → 03 Rest outside the mall → 04 Lie near the flowerbed → 05 Rest near the tea shop



Seek shade nearby to rest instead of going to the Union Service Station



Nearly half of the riders take no sun protection measures



Take a rest near the merchants



Sleeping by the flowerbed beside the shopping centre



Cooling off outside the shop

User Journey Map

	Waiting for the order		Delivering the order						Rest
User actions	Sit outside and wait for orders	Collect meals and goods on foot	Deliver orders by electric bike	View route and other order information on the phone	Encounter traffic jams and red lights	Walking into the residential compound	Lost in an unfamiliar neighborhood	Climbing stairs	Seek shade to rest
Feelings	I dare not stray too far, for fear of missing orders.	The goods are rather heavy. I'm terribly hot.	The sun is rather strong.	My phone is overheating and lagging. How can I view the navigation route?	The order deadline is approaching. Stopping makes me more anxious and hotter.	Blazing hot and sweltering. I carry the goods, puffed and panting.	I can't find my way, which makes me quite anxious.	Carrying goods up the stairs is exhausting; I'm sweating buckets	Resting helps, but I feel anxious knowing others are still earning
Mood	😐	😫	😫	😫	😫	😫	😫	😫	😐
Pain Point	Heat builds up due to lack of shade and airflow	Direct sunlight causes rapid physical exhaustion during prolonged activity	1. Overheated cushion causes discomfort 2. Rising body temperature	Lagging phone prolongs heat exposure and tires the user	Traffic jams raise heat and stress, boosting heatstroke risk	Extended deliveries elevate temperature and breathing strain	Heat and stress outdoors raise heatstroke risk	Climbing stairs in hot, cramped spaces strains the body's heat regulation	
Opportunities	Change into breathable work uniforms	Physiological monitoring for delivery completion	Increase guidance for tree-lined sections	Device to prevent phone overheating	Physiological monitoring for delivery completion	Work with property management to ease e-bike entry rules for delivery riders	Buildings and units should be clearly labeled	Physiological monitoring for delivery completion	

Persona

Xiaoming Wang



- Male
- **20 Year-old**
- Full-time
- Working Hour: 10

Other Circumstances

- Pay attention to the use of sunscreen products
- **Take appropriate breaks** when facing extreme conditions
- He doesn't push himself so hard to take on every order

" When it gets too hot, **find a shady spot to rest for a bit** before continuing deliveries."

" The hottest part of the day is midday, which also happens to be when the most orders come in. I have to **endure the sun to deliver orders.**

Dawei Liu



- Male
- **40 Year-old**
- Full-time
- Working Hour: 12

Other Circumstances

- Low awareness and compliance in sunscreen use
- Concerns about sunburn or tanning are minimal; **efficiency in taking orders is the priority**

"When waiting for orders, the sun beats down with nowhere to hide. **If you move elsewhere, you'll miss the order because others are nearby.**"

" If you can't handle the hardship of the heat, then that **order goes to someone else.**"

" Sun visors do block the sun, but wearing one makes you hotter—**better off without it.**"

Motivation



Stay healthy and avoid heatstroke

- They don't want to end up in the hospital and **miss out on earning money**
- They try to **stay a bit more comfortable** while delivering to avoid heat exhaustion

Pain Points



Lack of objective data for evaluating physical condition

- **No real-time physical data feedback** or high-temperature injury alerts
- Relying solely on subjective judgment to determine whether to continue working, **lacking objective criteria** to assess whether heatstroke has occurred

Maslow's Hierarchy of Needs



- Using delivery work as a stepping stone toward future career transitions

- Riders seek recognition through "top rider" titles or tips
- Confidence is undermined by unfair treatment
- Discrimination from security guards creates social exclusion
- Desire for customer understanding during delays

- Limited opportunities to rest during work
- **Exposure to extreme heat and sun, leading to risks of heatstroke and sunburn**
- Long-term occupational injuries (back, knees, wrists)
- Job instability

Competitor Analysis

Advantages		Disadvantages
New fabrics **(quick-dry, bamboo-textured ice silk, 3D honeycomb mesh)** offer excellent breathability	 Meituan	The cost of premium fabrics is rising, and it is **difficult for riders to procure them independently**
The striking red and gold colour scheme offers **high visual distinctiveness, enhancing brand recognition**	 JD.com	1. Unrevealed ventilation/ cooling design 2.The dark sections of the red-gold colour scheme **exhibit high heat absorption rates**
1.The material is **robust and durable**, resistant to damage and abrasion 2.With excellent **waterproof properties**	 Ele.me	The design of the waterproof coating **impedes breathability,** with insufficient attention paid to quick-drying properties and comfort.

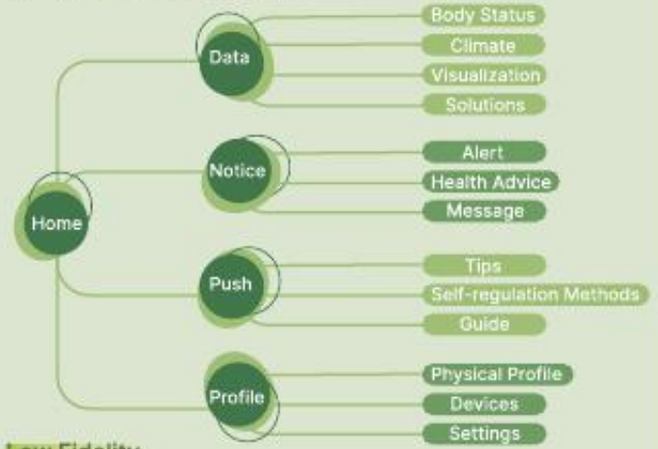
Concept



VitalRide is a product that **monitors the physiological data of delivery riders while providing cooling functionality.** Its purpose is to prevent fatal consequences arising from exertional heatstroke.

App Design

Information Architecture



Low Fidelity

Data



Notification



Alert



My





Adverts



User Test

- It's inconvenient to contact friends, and the user didn't know their online status.
- The information displayed on the device page is not detailed enough.
- Body data is displayed on different pages, making it inconvenient to view.

High Fidelity Interface

Colors

	507849
	C4E860
	D5F19F
	F7C537
	B56A4D
	FF9E3E
	F77E6D
	000000

Fonts

Figma Inter

Aa Medium

Aa Bold

System

IOS

Android



















Wearable Design

Sketches



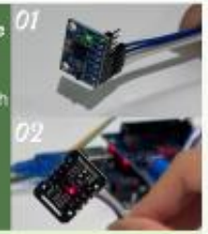
The wearable equipment contains two parts: a vest and a lightweight helmet, both equipped with **physiological data sensors and cooling mechanisms.** When used in conjunction with the VitalRide app, this system **monitors the user's health status and provides cooling.**

Technology Support

A PCM cooling ring in the vest's neckline cools the body by absorbing heat during phase change, powered by a removable battery.



01 **A posture sensor** is installed to **determine whether the rider has fallen** by detecting abnormal body tilt or sudden movement.
02 The vest incorporates **an ECG module** with silver-plated nano-mesh electrodes to **monitor heart rate** by detecting electrical signals from the skin.
03 **An inner-surface temperature sensor(1)** monitors body temperature.



The vest is made of **spandex-polyester** blend with moisture-wicking properties.

An inner-surface temperature sensor(2) monitors body temperature by **measuring skin contact heat.**



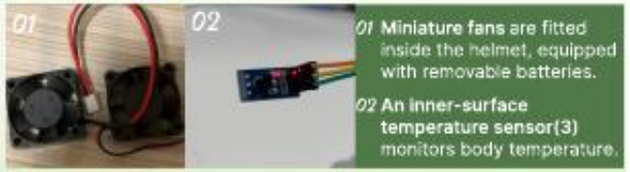
The underarm is made of **ice-silk bamboo-weave fabric.**

Product Modeling



Final Outcome





01 **Miniature fans** are fitted inside the helmet, equipped with removable batteries.
02 **An inner-surface temperature sensor(3)** monitors body temperature.

The helmet shell, made of **ABS+PC**, is lightweight and durable, with **multiple ventilation holes** and a **removable, washable inner lining.**



The vest features **fluorescent strips** that reflect light at night, enhancing rider safety.

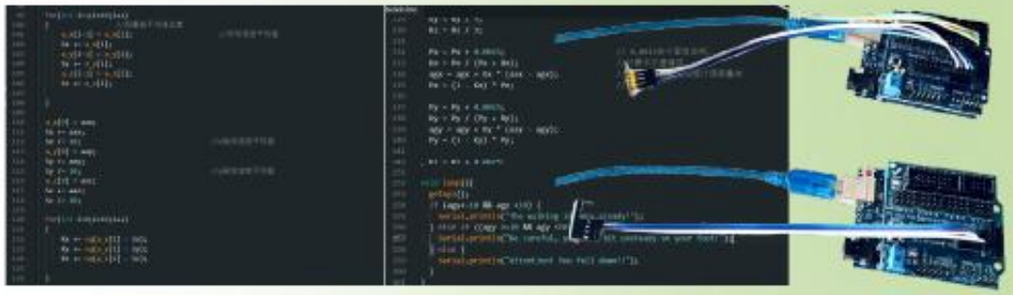
The vest features **breathable mesh panels** on the back and underarms to promote sweat evaporation.

Coding

Posture sensor



Temperature sensor and ECG module

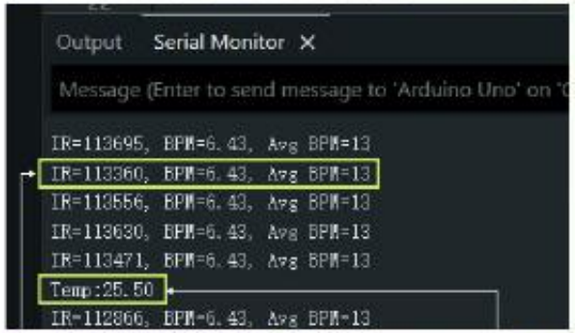


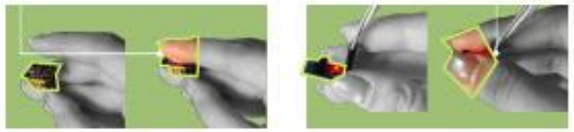
Usage Scenario



Testing







Final Storyboard



01 After collecting the order, the delivery rider sets off on his electric bike to deliver it.



02 His electric bike was stopped at the estate gate, so he had to deliver on foot.



03 On the way, he couldn't find his way and felt very hot , nervous and uncomfortable.



04 The mobile app alert sounded, warning him of elevated body temperature and accelerated heart rate.



05 He rested in the shade, opened the app, and activated the vest and helmet's cooling function.



06 Following a period of rest, the app indicated that his physiological data had returned to normal.